

Config Drift Detector

Project Reference Manual & Deployment Overview

1. Project Overview & Objective

The **Config Drift Detector (CDD)** is a Single-Page Web Application designed specifically for DevOps teams and software engineers. It facilitates instant alignment checks and security auditing by highlighting configuration drifts (mismatches) between environments (such as Development and Production) before they can impact active staging or live applications.

The application features a unique, premium **underwater ocean theme**. It utilizes glassmorphism overlays, bubble animations, and high-tech SVG visualizations to create an engaging user experience suited for both professional team presentations and project demos.

2. Workspace Directory Sizes & Storage Capacities

The project folder is organized according to modern React + Vite layouts. Below is the precise breakdown of disk space usage measured directly from the project directory:

Directory / Folder	Storage Size	Primary Purpose & Content
src/	0.09 MB (97,382 bytes)	Development Source. Houses all React code, visual components, SVG animations, stateful drift algorithms, and layout stylesheets.
public/	0.01 MB (14,553 bytes)	Static Assets. Contains baseline asset files and browser favicons loaded prior to JavaScript execution.
dist/	1.07 MB (1,117,668 bytes)	Production Build. Contains the compiled, minified, and optimized bundles generated via Vite. This optimized bundle is what is served to users on the live web.
node_modules/	160.73 MB (168,538,371 bytes)	Local Dependencies. Node packages installed locally (React, Tailwind CSS v4, js-yaml, jsPDF, Lucide React) required to build and run the development workspace.

3. Global Deployment & HTTPS Security Protocols

To allow your teammates to access the application, the production bundle (dist/) was published to a global Content Delivery Network (CDN) provided by **Surge.sh**.

Deployment Parameters:

- **Live Secure Link:** <https://strong-field.surge.sh/>
- **Secure Protocol:** HTTPS active. Encrypted using a wildcard TLS/SSL certificate provisioned automatically for Surge domains.
- **Availability:** Persistent. The page is hosted on public web servers and remains live 24/7/365, even if your local computer is off or if you close this workspace chat.

4. Core Features & Audit Capabilities

Teammates visiting the live site can interact with the following components:

- **Step-by-Step UI:** Upload or paste configuration files directly, click **Analyze Drift**, and inspect results instantly.
- **Dual-Pane Highlighting Code Viewer:** Side-by-side JSON panels highlighting modifications (yellow), additions (green), and deletions (red).
- **Executive Summary Metrics:** Live KPI metrics and a circular progress ring visualizing the alignment Health Score (84% on default sample).
- **Security Compliance Checklist:** Automated check-off validating parameters (SSL, CORS wildcards, debug logs, port mismatch).
- **Direct Exporters:** Downloads results as a structured CSV spreadsheet file or generates an offline PDF audit document in one click.